Team Turquoise, This is a reasonable first start, and, in a number of places, you have made good use of material that we have been discussing in class this semester. However, overall, I sense that you were focusing a lot more on the algorithms and "data science" stuff (data in → predictions out) and not as much on the formulation itself. I liked some of the ideas you introduced, but they often seemed divorced from the reality of banking and/or practical considerations of implementation.

For example, your proposal did not discuss any substantive strategies for dealing with missing data, though you had actually outlined earlier some of the relationships that might be useful elsewhere in the document. In addition, it seems that in some cases, you did not fully think through how your proposals would work in practice (e.g., how will you implement NLP sentiment analysis or how you are actually going to examine the impact of economic conditions,...) though you promised to do so. I appreciated your discussion of the model requirements, but you did not seem to connect that thinking to any discussion of how you would explain your model to bankers.

I also had the impression that you were making statements without foundation or support, particularly about banking and corporate behavior, and I also got the sense in some cases that your understanding of some of the approaches you recommended, and of credit modeling more generally, was fuzzy.

Proposal

Kiet Nguyen, Desmond Peng, Chloe Li

PLEASE INCLUDE TEAM NAME

This proposal would have benefitted from a more careful and deeper literature review. I noticed no finance references outside of those in the course text, and no references about banks or banking, etc., despite this being a huge literature in finance, and, less so, machine learning. Nonetheless, you made assertions, stated as fact, about what the bank would do and why.

Finally, I did not see any statements about how, if at all, you used AI, so I will assume that you are stating that you did not use AI tools for any part of this document. Overall, this felt very much like a standard data science proposal, rather than one that took a finance-oriented perspective. This was a good effort, but it still has a ways to go. Please consider reviewing the literature on statistical default prediction and also try to think through how you would implement — practically — some of the ideas you proposed.

Business Problem

Banca Massiccia bank lends to private, non-financial Italian firms. Every time a firm applies for a loan, a loan officer has to make two decisions: whether to lend at all, and, if so, what interest rate and underwriting fee to charge. Both decisions depend on one quantity: how likely the firm is to miss a principal or interest payment over the coming year.

How do these two differ?

Our problem statement: the bank needs a more accurate and better-calibrated estimate of each prospective borrower's probability of default (PD) over the next 12 months. The estimate must be built only from information available when the lending decision is made: the firm's most recent annual financial statement and basic firm characteristics (legal structure, industry, location).

We are improving the PD estimate specifically. The bank's expected loss (EL) on a loan can be written as PD × LGD × EAD, where LGD is the share of the loan lost after a default and EAD is the amount outstanding at default. Why are you telling me this?

Why this matters economically:

Mispricing loses money in both directions. If the bank underestimates a risky firm's PD, it charges too little and absorbs losses the rate never covered. If it overestimates a safe firm's PD, it charges too much and loses a profitable customer to a competitor. Either way, the bank leaves money on the table.

A weaker model makes the bank a target for adverse selection. Lending is competitive. When another bank can tell risky and safe borrowers apart better than Banca Massiccia can, the safe borrowers go to whichever bank offers them a better rate, and the risky ones end up at whichever bank underprices them. A bank with a weaker model therefore attracts a riskier book than it thinks it has. Stein (2005) shows that a model's ability to rank borrowers translates directly into lending profit through better cutoffs and pricing, so a better PD model is a profit improvement, not just a statistical one.

Data Mining

The modeling task will be a supervised one, meaning we will establish a labeled feature dataset where borrowers that defaulted in the past will be labeled as 1, otherwise they will be labeled as

0. Specifically, we will build a dataset based on some of the borrower data accumulated by Banca Massiccia, default or not. This dataset (referred to as "**the dataset**") will be used in the training, testing and validation process of the model and will be updated constantly when new applications come in, existing loans are paid, or when defaults occur.

1. **Eligibility of Borrower Samples**

   We should only have borrower samples in the dataset whose loans are already paid off or confirmed to be in default. This is because we cannot have "unlabeled" samples in the dataset for a supervised learning task, and for the borrowers whose loans are still outstanding, we are not sure about their loan status.

   Why?

   How does this relate to your stated business problem??

2. **Label and Feature Construction**

   Since we are predicting whether or not a borrower will default in the next 12 months, when constructing the dataset, we set the date of loan application as the benchmark date, and we label the borrower as 1 if they defaulted within 12 months of the benchmark date, otherwise we label them as 0. This prevents target leakage. The features will be constructed based on the data in the borrower's financial statements.

   What does this mean?

   Here, we made an important assumption that we only care about the probability of default in the first 12 months of application. If a borrower's financial standing changes so much after the first 12 months of application that we have to update the prediction, the data schema will need to be a rolling one. This might make sense given that the monitoring group will update the borrower's financial profile.

   The features, on the other hand, will be constructed from the financial statements provided by loan applicants. Aside from the financial metrics and ratios that can be calculated from the items on the financial statements, we could make use of the text in the explanatory notes and the management report by introducing NLP methods like sentiment analysis, topic modeling, and text embeddings.

   Are you promising to deliver this? How?

3. **Unit of analysis**

   The unit of analysis of this problem should be application, meaning each loan application will be evaluated independently. For example, the same company can apply for multiple loans throughout years and each application might be given a different verdict (approval/rejection) and different level of interest rate and underwriting fees. This does not mean that applications from the same company will not be jointly analyzed.

   What does this mean?

   Applicant profile and entity identity will be important features in our dataset.

   How?

   Another possible unit of analysis we considered but rejected is company / borrower, which means each company will be given one label at all times, and the label will be based on the most recent financial information of the company. We didn't choose this because we want to be able to go back to historical loan applications and audit previous decisions made by the bank.

   What does this mean?

   Are you promising to deliver this? How?

The above data schema is primarily for training and validation, and at inference, the data pipeline will work differently. After training (and retraining), the model's parameters will be set, and when new applications and corresponding financial statements come in, the feature pipeline will transform the raw text and numerical data into features and be fed into the model. We don't need labels at inference, and the outputs of the model are our desired outcome.

How?

Potential Features

Dependent Variable

Considering the late publication of financial reports, the definition of default in a specific year is unintuitive. We instead define the dependent variable as a **binary indicator**.
Taking the 2010 financial report as an example, the dependent variable equals 1 if the firm defaults within the 12 months after the financial report published (2011/07 - 2012/06 in this case), otherwise it is equal to 0.

How do you know this?

Independent Variable

We categorized the candidate features into several types, including business and structural risk, solvency and capital structure, liquidity, operating performance, and cash flow capacity. Most of these features were derived from existing features.

Considering the conventions in the financial sector, we take the following measures when building features:

- Use ratios: For example, cash / total assets, allowing comparisons between companies of different sizes. Size itself is represented separately using log(total assets).
- Collinearity: Many features are derived from common features, so we expected that some features are highly correlated. We will check features using VIF.

* feature_name represents an existing field in the dataset.

Feature Name	Definition	Note
<u>Business and Structural Risk</u>		
Industry	ateco_sector grouped into ~7 groups based on ATECO 2007 sections (How?)	The raw code has 62 categories, needs grouping
Legal structure	legal_struct (SRL, SRU, SPA, etc.), one-hot encoded	
Size	log(asst_tot)	

Solvency and Capital Structure		
Equity ratio	eqty_tot / asst_tot	
Debt ratio	(debt_st + debt_lt) / asst_tot	
Negative equity flag	1 if eqty_tot < 0	
Short-term debt share	debt_st / (debt_st + debt_lt)	
Bank debt concentration	(debt_bank_st + debt_bank_lt) / (debt_st + debt_lt)	~12% missing (short-form statements); needs imputation
Liquidity and Working Capital		
Cash ratio	cash_and_equiv / asst_tot	
Current assets ratio	asst_current / asst_tot	
Receivables ratio	AR / rev_operating	Replaces days receivables
Payables ratio	AP_st / asst_tot	~12% missing
Working capital ratio	wc_net / asst_tot	
Current ratio	asst_current / debt_st	Same as above
Profitability and Operating Performance		
ROA	roa	
EBITDA ratio	ebitda / asst_tot	
Operating margin	prof_operations / rev_operating	
Loss flag	1 if profit < 0	
Revenue, operating profit, net profit	Raw amounts	Not used directly, only used to compute the ratio above
ROE	roe	
Debt-Serving and Cash-Flow Capacity		
Interest coverage	ebitda / exp_financing	
Cash flow to debt	cf_operations / (debt_st + debt_lt)	
Debt / EBITDA	(debt_st + debt_lt) / ebitda	
Financial expenses	exp_financing	How do you plan to use this ?
Financial income	inc_financing	How do you plan to use this ?
Operating cash flow	cf_operations	Raw amount, not used directly
NLP Signal		

How, specifically, will you implement this?

Sentiment Analysis	Positive/Negative Tone in Annual Report Text
Overlapping Topics	If a company's report topic is similar to its peers, it means it faces systemic risks, idiosyncratic risk otherwise.

How to Validate the Model

Data Splitting Method: Split by Financial Year

We will use a walk-forward validation method:

Training Rounds	Training Data (Financial year)	Testing Data (Financial year)
1	2007 - 2008	2009
2	2007 - 2009	2010
3	2007 - 2010	2011 (part)
If the complete dataset includes more years, we may add more training rounds.		

Considering that in every year, financial reports are usually published between March and June of the following year, we must adopt a compromise forecast range.
Taking the 2010 financial report as an example, estimated to be published in June 2011, we can use this range to predict whether the company will default over the next 12 months (July 2011 - June 2012). The corresponding observation period for each year's financial report is as follows:

Financial Year	Observation Period	Included in Data
2007	2008/07 - 2009/06	Yes
……		
2010	2011/07 - 2012/06	Yes
2011	2012/07 - 2013/06	Only 2012/07 - 2012/12
2012	2013/07 - 2014/06	No

How, specifically, will tuning work?

- Hyperparameter tuning and feature selection are performed only within each training round. Data from the test year is not used for any estimation.
- Default data is only recorded up to 2012. If a default window of 6–18 months after 'stmt_date' is used, the 2011 column has only a partial observation period, and the 2012

Does this mean that each year's model will have different independent variables?

column has no observation period at all; therefore, these two years are not considered primary test years.

- The testing period covers the 2008–2009 financial crisis, so we can test the stability of the model under different economic conditions. How, specifically, will you do this?
- After validation, the final model will be reestimated using all available labeled data before being delivered.

Evaluate both "existing" and "new" companies

- Existing companies: Companies that appeared during training
- New companies: Companies that do not appear during training

The holdout dataset includes both kinds of companies. Considering that the model may perform better with familiar companies, we will split the test set into two groups and report them separately.

If the two sets perform differently, it may mean that the model relies on the company's specific features rather than general signals.

Evaluation Metrics

	Metrics	Purpose
Power	ROC curve, Accuracy ratio (CAP) How do these two differ? What additional information are you getting by using both?	Are defaulting companies classified as high risk?
Calibration	Calibration plot	Are the predicted PD values accurate?

Are there other more robust metrics?

How the model will be used

When an application arrives, the model takes the applicant's latest financial statement and returns a single PD. The bank uses that number in two different ways:

- **Approval:** origination uses the PD *ranking* to decide which applicants fall above or below the bank's acceptance cutoff. How do you know that the bank uses a cut-off?
- **Pricing:** origination uses the PD *level* to set a rate and fee that cover expected loss plus a margin.

The risk management group can use the same estimates to monitor the portfolio over time.

What a successful model must do

Because the PD is used both for ranking and for pricing, the model has to perform well on two separate dimensions:

- **Discriminatory power:** firms that go on to default should receive higher PDs than firms that don't. This is what improves approval decisions.
- **Calibration:** the PD levels should match reality. Among firms assigned a 3% PD, about 3% should actually default. This is what makes pricing correct. A model can rank well and still be badly calibrated, which would lead to systematically mispriced loans.

The model must also hold up in the conditions it will actually face:

- **A future time period.** The model will be used for years it was not trained on, so we validate it out-of-time (see the validation section). The data covers the 2008–09 financial crisis and the 2011–12 Italian sovereign debt crisis, which lets us check stability across very different credit environments.
- **Borrowers the bank has never seen.** New applicants have no history with the bank, so the model cannot depend on firm identity.
- **Credit officers' judgment.** Its main drivers should be ones a credit officer recognizes, such as leverage, profitability, liquidity, and debt-service capacity. A model officers do not trust will not be used.

Why we use a finance-informed formulation

These are private firms with no traded equity, so market-based default models that rely on stock prices (Merton-style structural models) do not apply. Default prediction for private firms has instead been built on accounting ratios. This line of work runs from Altman's Z-score (1968) and Ohlson's model (1980) to Moody's RiskCalc for private companies (Falkenstein, Boral & Carty, 2000), the industry standard for this exact problem. We build on that tradition in three ways:

- **Features:** we construct ratios with established economic meaning rather than feeding raw balance-sheet amounts to an algorithm.
- **Timing:** we respect the lag between a fiscal year-end and when its statement actually becomes available to the bank, so the model never uses information the bank would not have had.
- **Evaluation:** we evaluate on power and calibration separately, because the bank uses the PD for both.

Scope and limitations

- **Population:** the model covers non-financial firms with more than €1.5MM in assets, matching the bank's data.
- **Horizon:** it predicts default within 12 months of the point at which the bank could act on the financial statement. How do you know this?
- **Selection bias:** the data includes only firms the bank actually lent to. Rejected applicants never appear, so the model learns from an already-screened population. Its PDs may be less reliable for applicants who look very different from past borrowers.

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